TICCO ROSS

P-800

8-voice programmable synthesizer · AU / VST3

User manual · version 0.1



A reconstruction of a classic 1980s digital-analog hybrid: eight voices of stacked square-wave footages, three six-stage envelopes, one shared resonant filter and a stereo chorus.

1. Getting started

Installing

- Run the **P-800 .pkg** installer. It installs the Audio Unit to */Library/Audio/Plug-Ins/Components* and the VST3 to */Library/Audio/Plug-Ins/VST3*.
- The installer is not yet signed by Apple. If macOS blocks it, right-click the .pkg and choose **Open**, or allow it under **System Settings › Privacy & Security**.
- In Ableton Live, P-800 appears under **Plug-ins › Audio Units › Ticco Ross** (or VST3). Rescan plug-ins if it does not show up.
- P-800 runs at any sample rate. Below 88 kHz it runs its engine at twice the host rate internally, so it sounds the same at 44.1, 48 and 96 kHz.

First sounds

P-800 opens on factory program **11 Strings**. Step through the 64 factory programs with the **< PROG** and **PROG >** buttons under the display, from your host's preset menu, or by typing a program number on the keypad (section 3).

2. The panel



Section	Controls
Joystick / Tune	**Tune** slider (±100 cents, double-click to reset). **Joystick**: left/right bends pitch, up adds vibrato (MG to DCO), down adds filter wobble (MG to VCF). It springs back when released.
Volume	Master output level.
Key assign	**POLY**: normal playing. **HOLD**: notes keep sounding after you let go, until you press POLY.
Programmer	LED display (program, parameter, value), keypad, PROG/PARA key, **DOWN** / **UP**.
Performance	**VOICES** switch (8 voices WHOLE / 4 voices DOUBLE, same as P18) and **CHORUS** on/off (P48).
Parameter chart	Every sound parameter, laid out like the printed chart on the original. Each cell shows its name, number, range and current value.

3. Editing sounds

The original instrument has no knobs: you pick a parameter by number and change its value. P-800 keeps that workflow and adds direct mouse editing on the chart. All methods change the same parameters.

On the chart

- **Click** a cell to select it. Its number and value appear on the LED display.
- **Drag** a cell up or down to change its value, or use the **scroll wheel** over it.
- Press **UP** / **DOWN** to step the selected parameter one value at a time.

With the keypad

- The keypad has the digits 1-8, as on the original, plus the **PROG / PARA** key.
- In **PARA** mode, type two digits to select a parameter: **4** then **1** selects P41 Cutoff.
- Press **PROG/PARA** to switch to **PROG** mode. Two digits now load a factory program: **2** then **4** loads 24 “1999”.
- After the first digit the display shows it with a dash, waiting for the second.

Saving your sounds

Your edits are stored with your Live set like any plug-in. To keep a sound for other projects, save it as a preset in your host (in Live: the save button in the device title bar). Loading a factory program replaces the current edits.

4. Playing

Joystick and controllers

- **Joystick X** bends pitch by up to ±**Bend Range** semitones (1-12, default 2; set it from your host's parameter list).
- **Joystick up** adds MG vibrato to the DCOs; **down** adds MG movement to the filter. Both add to the program's own MG depths (P83/P84) and follow the MG speed (P81).
- Your keyboard's **pitch wheel** adds to joystick X, and its **mod wheel** (CC1) works like pushing the joystick up.
- Joystick X and Y are host parameters, so you can automate them.

Voices and the shared filter

WHOLE mode gives 8 voices with DCO 1 only; DOUBLE mode gives 4 voices, each with DCO 1 and DCO 2 for fatter, detuned or interval sounds. All voices go through **one shared filter**, as on the original, so a new note can retrigger the filter envelope for chords already sounding (see P46 Trigger). Keyboard tracking follows the highest note held.

5. Parameter reference

DCO 1

No.	Name	Range	What it does
11	Octave	1-3	Pitch register of DCO 1: 1 low (16'), 2 middle, 3 high.
12	Waveform	1-2	1 square. 2 the stepped saw-like mix (footages weighted 1, 1/2, 1/4, 1/8).
13-16	16' / 8' / 4' / 2'	0-1	Switch each square-wave footage on or off. Combine them to build the tone.
17	Level	0-31	DCO 1 volume.
18	Mode	1-2	1 WHOLE: 8 voices, DCO 1 only. 2 DOUBLE: 4 voices, DCO 1 + DCO 2.

DCO 2 (heard in DOUBLE mode)

No.	Name	Range	What it does
21	Octave	1-3	Pitch register of DCO 2.
22	Waveform	1-2	As P12.
23-26	16' / 8' / 4' / 2'	0-1	DCO 2 footages.
27	Level	0-31	DCO 2 volume.
31	Interval	0-12	DCO 2 transposition in semitones above DCO 1.
32	Detune	0-3	Detunes DCO 2 downward, up to about -20 cents.

Noise

No.	Name	Range	What it does
33	Noise level	0-15	White noise into the filter, shaped by DEG 3.

VCF (one filter shared by all voices)

No.	Name	Range	What it does
41	Cutoff	0-99	Filter cutoff frequency.
42	Resonance	0-15	Emphasis at the cutoff. Very high settings ring and whistle.
43	Keyboard track	0-2	0 off, 1 half, 2 full: cutoff follows the highest note played.
44	EG polarity	1-2	1 inverted, 2 normal direction of the DEG 3 sweep.
45	EG intensity	0-15	How far DEG 3 sweeps the cutoff.
46	Trigger	1-2	1 single: DEG 3 restarts only when all keys are released. 2 multi: every note restarts it.
48	Chorus	0-1	Stereo BBD-style chorus on or off.

DEG 1 / DEG 2 / DEG 3 (six-stage envelopes)

No.	Name	Range	What it does
51-56	DEG 1	0-31	Volume envelope of DCO 1: Attack, Decay, Break point, Slope, Sustain, Release.
61-66	DEG 2	0-31	Volume envelope of DCO 2, same stages.
71-76	DEG 3	0-31	Filter and noise envelope, same stages.

MG (modulation generator)

No.	Name	Range	What it does
81	Frequency	0-15	MG speed (16 steps from the original ROM table).
82	Delay	0-15	Time before the MG fades in after a note starts.
83	DCO depth	0-15	Vibrato depth, up to about +/-160 cents.
84	VCF depth	0-15	Filter modulation depth.

Envelope stages (DEG 1-3)

Each DEG rises at the **Attack** rate to full level, falls at the **Decay** rate to the **Break point** level, then moves at the **Slope** rate toward the **Sustain** level, where it stays while the key is held. On release it falls at the **Release** rate. The break point and slope give the two-part decays typical of this instrument.

Host-only parameter	Range	What it does
Volume	0-1	Master output (the panel fader).
Tune	±100 cents	Fine tuning (the panel slider).
Joystick X / Joystick Y	-1 to +1	The panel joystick; automatable.
Bend Range	1-12 st	How far joystick X and the pitch wheel bend.
Hold	off / on	Key assign HOLD.

6. Factory programs

The 64 original factory programs, decoded from a preserved factory cassette. Numbers are bank (1-8) and program (1-8), as on the original.

11	Strings	**51**	Synthenelli
12	Acoustic Piano	**52**	Synthe Flute/Brass
13	Synthe Brass	**53**	Synthicord I
14	Flute with Chiff	**54**	Synthicord II
15	Resonant Sweep	**55**	Saxo-Brass
16	Bowed Cellos	**56**	Square Wave Ensemble
17	Full Organ	**57**	Chime Piano
18	Fat Synthe	**58**	Pipe Organ
21	Percussive Synthe	**61**	Bowed Violas
22	Muted Brass	**62**	Digi-Brass
23	Square Wave Lead	**63**	Organ Upper Manual
24	1999	**64**	Obi Brass
25	Low Strings	**65**	Soft Piano
26	Percussive Synthe (2)	**66**	Reverse Sweep
27	Bright Perc. Organ	**67**	Accordion
28	Science Bass	**68**	Steel Drums
31	Lambda	**71**	8-Note Brass
32	OB Cross	**72**	Honky Keys
33	High Strings	**73**	Organ Lower Manual
34	Double Crescendo Bass	**74**	8-Note Strings
35	Synthe Bass I	**75**	Reso Synthe
36	Reverse Bowed Synthe	**76**	Echo Synthe
37	Jazz Guitar	**77**	Black Hole
38	Helicopter	**78**	Trilling Pipes
41	Double Tongued Bass	**81**	Brass in 5ths
42	Synful Eyes	**82**	Percussive Pipe Organ
43	Wet Ones	**83**	Lunar Flutes
44	Science Strings	**84**	Synthe Bass III
45	Synthe Bass II	**85**	Vibes
46	Crescendo Brass	**86**	Low Flute
47	Chorus Piano	**87**	Who Me?
48	Jet Plane	**88**	UFO Swoop

7. About P-800

P-800 is built on **ISLA Poly-800** by Interspock, an open-source reconstruction of the Korg Poly-800 MkI / EX-800 that recovers its modulation generator, envelope tables and factory data from the original firmware and cassette. The filter and parts of the signal path derive from Nick Copeland's Bristol synthesizer emulator.

Differences from the reconstruction it is built on

- Runs internally at 88.2/96 kHz at lower host rates, because the engine's lower-rate filter branch becomes unstable at full resonance.
- Adds joystick bend and modulation, HOLD, and the hardware-style panel.
- Not included: the original's sequencer and chord memory (P86-P88 and KEY ASSIGN CHORD), and cassette or MIDI SysEx program transfer.

Licence

P-800 is free software under the **GNU General Public License v3.0 or later**. You may use, study, share and change it under that licence. The complete source code is at **github.com/Spunkmeyer77/Poly800**.

Korg and Poly-800 are trademarks of Korg Inc. P-800 is an independent project and is not affiliated with or endorsed by Korg.